

PYTHON FOR BEGINNERS · DAY 10

Final Project



Putting it all together

Instructor: Homaira Mohammadi

- 🛡️ Handle errors with **try / except**
- 📝 Plan a project like a programmer
- 🎓 Build a Student Grade Manager
- 🎯 Choose and build **your own** project

LOOK HOW FAR YOU'VE COME 🗺️

Everything you've learned

1 print() & comments	**2** variables, input & operators	**3** if / elif / else
4 while loops	**5** for loops & range()	**6** strings
7 lists	**8** functions	**9** dictionaries & tuples

That's enough to build **real programs**! 💪

ONE LAST PROBLEM

What if the user types “twenty”?

```
age = int(input("Age: "))
```

```
Age: twenty
ValueError: invalid literal for int()
```

The program **crashes** 💥

Real users make mistakes all the time.

We need a way to **catch** the error and ask again.

TRY / EXCEPT 🛡️

Catch the error instead of crashing

Code

```
text = "twenty"
try:
    age = int(text)
    print("Age:", age)
except ValueError:
    print("Not a number!")
```

Output

```
Not a number!
```

try: code that might fail
except: what to do if it fails

A HELPER FOR EVERY PROJECT

Keep asking until it’s a number

```
def get_int(prompt):
    while True:
        try:
            return int(input(prompt))
        except ValueError:
            print("Please type a number.")

age = get_int("Age: ")
```

```
Age: twenty
Please type a number.
Age: 20
```

Combines a **function**, a **while** loop and **try / except**.

THINK BEFORE YOU TYPE 📝

How programmers plan a project

1

Understand

What should it do? Who uses it?

2

Plan the data

Which lists and dictionaries?

3

Split into jobs

Each job becomes a function

4

Write pseudocode

The steps in plain words

5

Build & test

One small piece at a time

6

Improve

Handle bad input, tidy output

🎓 OUR PROJECT

Student Grade Manager

The data: name → list of scores

```
students = {
    "Sara": [88, 61],
    "Omar": [95],
}
```

The jobs: one function each

Function	Job
average(scores)	average of a list
get_grade(avg)	A, B, C, D or F
add_student()	add a new student
add_score()	save a score
show_report()	print the table
main()	the menu loop

🚀 BRING IT TOGETHER

The menu: main()

```
def main():
    students = {}
    while True:
        choice = input("Choose (1-5): ")
        if choice == "1":
            add_student(students)
        elif choice == "2":
            add_score(students)
        elif choice == "3":
            show_report(students)
        elif choice == "5":
            break
```

```
Name    Avg   Grade
Sara    74.5  B
Omar    95.0  A
```

A real program, built from **everything** you learned. 🎉

💡 THE GOLDEN RULE

Build small. Test often.

Write a few lines, run them, fix them — then add the next piece.

🎯 YOUR FINAL PROJECT

Choose one — or invent your own

✊

Rock, Paper, Scissors

Best of 3 · ⭐ ⭐

💸

Expense Tracker

Totals by category · ⭐ ⭐

🔤

Hangman

Guess the word · ⭐ ⭐ ⭐

🏦

Bank Account

Deposit & withdraw · ⭐ ⭐

🧠

Quiz Game

Questions & score · ⭐ ⭐

💡

Your own idea!

Check it with your teacher

✅ PROJECT CHECKLIST

Your project should have…

- ✓ Variables + a list or dictionary
- ✓ At least one loop
- ✓ No crash on bad input
- ✓ Clear, friendly output
- ✓ if / elif / else
- ✓ At least two of your own functions
- ✓ Comments that explain the code

🎤 PRESENTING YOUR PROJECT

4 things to show (3–5 minutes)

1

What does your program do?

2

Show it running — a live demo

3

Code you're proud of

4

What was hard, and how you solved it

🚀 WHAT'S NEXT?

Keep learning

Topic	What it lets you do
Files — open()	save and load data
Modules — import	use ready-made tools like math and random
Classes (OOP)	create your own data types
turtle / pygame	draw graphics and make games
pandas / matplotlib	analyse data and draw charts

Free practice links are in **resources/next-steps.md** on GitHub.



Well done, programmer! 🎉

You finished Python for Beginners.

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